SIMATS SCHOOL OF ENGINEERING

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Assignment 4

1. Give a FOL translations for the following sentences: (10M) (CO4) (BLT4)
   a. Vegetarians do not eat meat.
   b. Not all vegetarians eat eggs.

2. Give a set of facts and inferences necessary to prove the following assertions: (10M) (CO4) (BLT4)
   a. McDonalds is not a vegetarian restaurant.
   b. Some vegetarians can eat at McDonalds.

Don't just place these facts in your knowledge base. Show that they can be inferred from some more general facts about vegetarians and McDonalds.

3. Give FOL translations for the following sentences that capture the temporal relationships between the events. (10M) (CO4) (BLT4)
   a. When Mary's flight departed, I ate lunch.
   b. When Mary's flight departed, I had eaten lunch.

4. The following rule, which we gave as a translation, is not a reasonable definition of what it means to be a vegetarian restaurant. (10M) (CO4) (BLT4)

$\forall x \, VegetarianRestaurant(x) \Rightarrow Serves(x, VegetarianFood)$

Answer 1. FOL translations

a. Vegetarians do not eat meat.

$\forall x \, \forall y \, (Vegetarian(x) \wedge Meat(y) \Rightarrow \neg Eats(x,y))$

Event form: $\forall x \, (Vegetarian(x) \Rightarrow \neg \exists e \, \exists y \, (Eating(e) \wedge Eater(e,x) \wedge Eaten(e,y) \wedge Meat(y)))$

Every vegetarian x and every piece of meat y are such that x does not eat y; equivalently, there is no eating event in which a vegetarian eats meat.

b. Not all vegetarians eat eggs.

$\neg \forall x \, (Vegetarian(x) \Rightarrow \exists y \, (Egg(y) \wedge Eats(x,y)))$

Equivalent: $\exists x \, (Vegetarian(x) \wedge \neg \exists y \, (Egg(y) \wedge Eats(x,y)))$

Pushing the negation inward ($\neg\forall$ becomes $\exists\neg$, and $\neg(A \Rightarrow B)$ becomes $A \wedge \neg B$) shows that there is at least one vegetarian who eats no eggs. This is weaker than "no vegetarian eats eggs", which would be a universal negation.

Answer 2. Facts and inferences

Instead of asserting the conclusions, a few general facts about vegetarians, food and restaurants are placed in the knowledge base, and both assertions are derived from them.

General facts (rules):

R1 $\forall x \, (VegetarianRestaurant(x) \Rightarrow Restaurant(x) \wedge \forall y \, (Serves(x,y) \Rightarrow VegetarianFood(y)))$

R2 $\forall y \, (VegetarianFood(y) \Leftrightarrow \neg ContainsMeat(y))$

R3 $\forall y \, (Hamburger(y) \Rightarrow ContainsMeat(y))$

R4 $\forall y \, (Salad(y) \Rightarrow \neg ContainsMeat(y))$

R5 $\forall x \, \forall y \, (Vegetarian(x) \wedge VegetarianFood(y) \Rightarrow CanEat(x,y))$

R6 $\forall x \, \forall r \, \forall y \, (Serves(r,y) \wedge CanEat(x,y) \Rightarrow CanEatAt(x,r))$

Specific facts about McDonalds:

F1 $Restaurant(McDonalds)$

F2 $Serves(McDonalds, BigMac)$

F3 $Hamburger(BigMac)$

F4 $Serves(McDonalds, SideSalad)$

F5 $Salad(SideSalad)$

F6 $Vegetarian(Mary)$

a. McDonalds is not a vegetarian restaurant.

#	Statement	Justification
1	$Hamburger(BigMac)$	F3
2	$ContainsMeat(BigMac)$	R3, universal instantiation + modus ponens with 1
3	$\neg VegetarianFood(BigMac)$	R2 ($\Leftarrow$ read left to right in contrapositive) with 2
4	$Serves(McDonalds, BigMac)$	F2

#	Statement	Justification
5	*Serves*(*McDonalds*,*BigMac*) ∧ ¬*VegetarianFood*(*BigMac*)	∧-introduction, 3 and 4
6	¬∀*y* (*Serves*(*McDonalds*,*y*) ⇒ *VegetarianFood*(*y*))	from 5 (a counterexample y = BigMac)
7	*VegetarianRestaurant*(*McDonalds*) ⇒ ∀*y* (*Serves*(*McDonalds*,*y*) ⇒ *VegetarianFood*(*y*))	R1, universal instantiation, ∧-elimination
8	¬*VegetarianRestaurant*(*McDonalds*)	modus tollens, 6 and 7

b. Some vegetarians can eat at McDonalds.

#	Statement	Justification
1	*Salad*(*SideSalad*)	F5
2	¬*ContainsMeat*(*SideSalad*)	R4 with 1 (modus ponens)
3	*VegetarianFood*(*SideSalad*)	R2 (⇐ direction) with 2
4	*Vegetarian*(*Mary*) ∧ *VegetarianFood*(*SideSalad*)	F6, 3, ∧-introduction
5	*CanEat*(*Mary*, *SideSalad*)	R5 with 4 (modus ponens)
6	*Serves*(*McDonalds*, *SideSalad*) ∧ *CanEat*(*Mary*, *SideSalad*)	F4, 5, ∧-introduction
7	*CanEatAt*(*Mary*, *McDonalds*)	R6 with 6 (modus ponens)
8	∃*x* (*Vegetarian*(*x*) ∧ *CanEatAt*(*x*, *McDonalds*))	F6, 7, existential generalisation (x = Mary)

Neither conclusion was stored directly. (a) follows from general facts about meat, hamburgers and what a vegetarian restaurant must serve, and (b) follows from the facts that vegetarians can eat vegetarian food and that a person can eat at any place that serves something they can eat.

Answer 3. Temporal relationships between events

Each sentence is represented with event variables and explicit time intervals (Reichenbach-style). Eating(e) and Departing(e) are event types, i_e is the interval of event e, Now is the time of speaking, Precedes(i,j) means interval i ends before j begins, and Overlaps(i,j) means the two intervals coincide in time.

a. When Mary's flight departed, I ate lunch.

∃*e1* ∃*e2* ∃*f* ∃*l* ∃*i* ∃*j* (*Flight*(*f*) ∧ *FlightOf*(*f*,*Mary*) ∧ *Departing*(*e1*) ∧ *Departer*(*e1*,*f*) ∧ *IntervalOf*(*e1*,*i*) ∧ *Lunch*(*l*) ∧ *Eating*(*e2*) ∧ *Eater*(*e2*,*Speaker*) ∧ *Eaten*(*e2*,*l*) ∧ *IntervalOf*(*e2*,*j*) ∧ *Overlaps*(*i*,*j*) ∧ *Precedes*(*i*,*Now*) ∧ *Precedes*(*j*,*Now*))

Both events are in the past (simple past), and the eating interval j coincides with the departure interval i: the speaker was eating lunch at the time the flight departed.

b. When Mary's flight departed, I had eaten lunch.

∃*e1* ∃*e2* ∃*f* ∃*l* ∃*i* ∃*j* (*Flight*(*f*) ∧ *FlightOf*(*f*,*Mary*) ∧ *Departing*(*e1*) ∧ *Departer*(*e1*,*f*) ∧ *IntervalOf*(*e1*,*i*) ∧ *Lunch*(*l*) ∧ *Eating*(*e2*) ∧ *Eater*(*e2*,*Speaker*) ∧ *Eaten*(*e2*,*l*) ∧ *IntervalOf*(*e2*,*j*) ∧ *Precedes*(*j*,*i*) ∧ *Precedes*(*i*,*Now*))

The past perfect ("had eaten") places the eating interval j completely *before* the departure interval i, and the departure itself is in the past. In Reichenbach's terms, (a) has event time E = reference time R, whereas (b) has E < R < speech time S. The only difference between the two formulas is Overlaps(i,j) versus Precedes(j,i).

Answer 4. Why the rule is not a good definition, and a better one

Rule given: $\forall x\ VegetarianRestaurant(x) \Rightarrow Serves(x, VegetarianFood)$

Problems with the rule:

1. It is only a one-way (necessary) condition. It says a vegetarian restaurant serves vegetarian food, but a restaurant that serves nothing but vegetarian food cannot be proved to be a vegetarian restaurant from it.
2. It is too weak. “Serves vegetarian food” is satisfied by any restaurant with a single vegetarian dish, so McDonalds (which serves a salad) would pass, although it also serves hamburgers. A vegetarian restaurant should serve *only* vegetarian food.
3. VegetarianFood is used as a single object (a constant), not as a category of dishes, so the rule cannot talk about individual dishes such as a particular salad or burger.
4. It does not say that the thing is a restaurant at all.

Better definition: a vegetarian restaurant is a restaurant that serves something, and everything it serves is vegetarian food.

$$\forall x\ (VegetarianRestaurant(x) \Leftrightarrow Restaurant(x) \wedge \exists y\ Serves(x,y) \wedge \forall y\ (Serves(x,y) \Rightarrow VegetarianFood(y)))$$

This is a two-way definition ($\Leftrightarrow$), so it both lets us conclude that a restaurant is vegetarian when all its dishes are vegetarian and lets us rule it out when it serves even one non-vegetarian dish, as in Answer 2(a). The clause $\exists$y Serves(x,y) stops an empty restaurant that serves nothing from counting vacuously as vegetarian.